Popularity-Adaptive Kinetic Model

Albi, Calzola & Dimarco (2025) — data-driven; Albi, Calzola & Piu (2025) — controlled

Agent state: (v, c) where v ∈ [–1,1] is opinion, c ≥ 0 is popularity (# followers)

Contact Dynamics

$$c' = c - \Psi_\delta(c/\bar{c})\, c + \eta\, c$$

Ψ_δ: prospect-theory value function
c̄: reference satisfaction level
η: random perturbation
Low-follower accounts gain followers more easily than high-follower accounts lose them.

Opinion Dynamics

$$v' = v + \gamma P(v, v_*, c, c_*)(v_* - v) + D(v)\,\xi$$

P = H · K:
H: bounded confidence (interact only if $|v - v^*| < \delta$)
$K = c^{*p}/(c^p + c^{*p})$: popularity weight
Agents update more toward high-popularity peers.

Bidirectional Coupling (controlled version)

$$\Phi(v) = \theta[(v - m_v)^2 - \delta_\Phi^2]$$

Opinion feeds back into contact dynamics. Minority opinions (far from m_v) lose followers; agents near the mean gain them → echo chambers.

Optimal Control Laws

Contact Control κ*

Counteracts minority-opinion agents losing followers:

$$\kappa^* = \frac{\lambda\beta}{\gamma_c} R_c(c)\, H_c(v)$$

R_c(c): sigmoid activating when popularity < c_min
H_c(v): sigmoid activating when local opinion density > ρ*

Opinion Control u*

Steers agents toward target opinion ṽ:

$$u^* = -\frac{R_v(c)\, H_v(\rho)\, [v + \alpha P(v_* - v) - \tilde{v}]}{\gamma_v + \alpha\, R_v(c)\, H_v(\rho)}$$

R_v(c): activates based on contact level
H_v(ρ): activates based on local opinion density
γ_v: regularization parameter

Controlled Fokker-Planck Equation for joint density f(v, c, t):

$$\partial_t f = -\alpha\, \partial_v\Big[\Big(\mathcal{P}[f] - \frac{R_v H_v(v - \tilde{v})}{\gamma_v}\Big) f\Big]$$

← *opinion drift + opinion control*

$$-\beta\, \partial_c\Big[\Big(\tfrac{\mu}{2}\ln\tfrac{c}{\bar{c}} + \Phi(v) - \tfrac{\lambda\beta}{\gamma_c} R_c H_c\Big) c\, f\Big]$$

← *contact drift + Φ(v) feedback + contact control*

$$+\tfrac{\sigma^2}{2}\partial_{vv}(D(v)^2 f) + \tfrac{\nu^2}{2}\partial_{cc}(c^2 f)$$

← *opinion diffusion + contact diffusion*

Implemented via Nanbu-type Monte Carlo particle scheme in the quasi-invariant regime.

Popularity-Adaptive Control Model

Control of Contact

Agent Dynamics to Mean-Field Model

Algorithm 1 Asymptotic particle-based scheme (Nanbu-like algorithm)

Fix $0 < \varepsilon = \Delta t < 1$ and N_s.
Sample $\{v_i^0, c_i^0\}_{i=1}^{N_s}$ from the initial distribution $f_0(v, c)$.
for $n = 0 : N_t - 1$ **do**
  Set $N_c = \text{round}(N_s/2)$.
  Select N_c unordered random pairs (i, i_*) uniformly without repetition.
  for $k = 1 : N_c$ **do**
    Let (i, i_*) be the k-th selected pair.
    Sample noise terms $\eta^n_{\varepsilon,i}, \eta^n_{\varepsilon,i_*}$ with $\langle \eta \rangle = 0$, $\langle \eta^2 \rangle = \varepsilon \nu^2$, and $\xi^n_i, \xi^n_{i_*} \sim \mathcal{N}(0,1)$.
    Compute the contact-control contributions

$$\kappa^n_j = \lambda \frac{\beta}{\gamma_c} R_c(c^n_j) H_c(v^n_j), \qquad j \in \{i, i_*\}.$$

    Compute the opinion controls

$$u^n_i = -\frac{R_v(c^n_i) H_v(v^n_i) \left(v^n_i + \varepsilon\alpha (P(v^n_i, v^n_{i_*}, c^n_i, c^n_{i_*})(v^n_{i_*} - v^n_i)) - \tilde{v} \right)}{\gamma_v + \varepsilon\alpha R_v(c^n_i) H_v(v^n_i)},$$

$$u^n_{i_*} = -\frac{R_v(c^n_{i_*}) H_v(v^n_{i_*}) \left(v^n_{i_*} + \varepsilon\alpha (P(v^n_{i_*}, v^n_i, c^n_{i_*}, c^n_i)(v^n_i - v^n_{i_*})) - \tilde{v} \right)}{\gamma_v + \varepsilon\alpha R_v(c^n_{i_*}) H_v(v^n_{i_*})}.$$

    Update contacts

$$c^{n+1}_j = c^n_j - \varepsilon\beta \left(\Psi^\varepsilon \left(\frac{c^n_j}{\bar{c}} \right) + \Phi(v^n_j) - \kappa^n_j \right) c^n_j + \eta^n_{\varepsilon,j}\, c^n_j, \qquad j \in \{i, i_*\}.$$

    Update opinions

$$v^{n+1}_i = v^n_i + \varepsilon\alpha \left(P(v^n_i, v^n_{i_*}, c^n_i, c^n_{i_*}) + u^n_i \right) + \sqrt{\varepsilon}\, \sigma\, D(v^n_i)\, \xi^n_i,$$

$$v^{n+1}_{i_*} = v^n_{i_*} + \varepsilon\alpha \left(P(v^n_{i_*}, v^n_i, c^n_{i_*}, c^n_i) + u^n_{i_*} \right) + \sqrt{\varepsilon}\, \sigma\, D(v^n_{i_*})\, \xi^n_{i_*}.$$

  end for
end for

Experiment Results

Numerical Experiments Setup

Effect of Contact vs Opinion Control

Competing Groups and Polarization

Emergence of Echo Chambers